

# 05. Database table and its data types

## Create table

```
CREATE TABLE employees (  
  id SERIAL PRIMARY KEY,  
  name VARCHAR(100),  
  designation VARCHAR(100),  
  salary NUMERIC(10,2)  
);
```

| Category       | Data Type        | Description          | Example                | Range / Notes    |
|----------------|------------------|----------------------|------------------------|------------------|
| Integer        | SMALLINT         | Small integer        | age SMALLINT           | ±32K             |
|                | INTEGER / INT    | Standard integer     | id INT                 | ±2.1B            |
|                | BIGINT           | Large integer        | views BIGINT           | Very large       |
| Auto Increment | SERIAL           | Auto increment int   | id SERIAL              | Creates sequence |
|                | BIGSERIAL        | Large auto increment | id BIGSERIAL           | 64-bit           |
| Exact Number   | NUMERIC(p,s)     | Exact precision      | price NUMERIC(10,2)    | Financial data   |
|                | DECIMAL          | Same as NUMERIC      | amount DECIMAL         | Exact            |
| Floating       | REAL             | Single precision     | rating REAL            | Approx           |
|                | DOUBLE PRECISION | Double precision     | temp DOUBLE PRECISION  | Approx           |
| Text           | CHAR(n)          | Fixed length         | gender CHAR(1)         |                  |
|                | VARCHAR(n)       | Variable length      | name VARCHAR(100)      |                  |
|                | TEXT             | Unlimited text       | description TEXT       | Most used        |
| Boolean        | BOOLEAN          | True/False           | is_active BOOLEAN      |                  |
| Date/Time      | DATE             | Date only            | dob DATE               |                  |
|                | TIME             | Time only            | start_time TIME        |                  |
|                | TIMESTAMP        | Date + time          | created_at TIMESTAMP   |                  |
|                | TIMESTAMPTZ      | Timestamp with TZ    | created_at TIMESTAMPTZ | Recommended      |
| Duration       | INTERVAL         | Time duration        | duration INTERVAL      |                  |
| JSON           | JSON             | JSON text            | data JSON              |                  |
|                | JSONB            | Binary JSON          | data JSONB             | Faster queries   |
| Binary         | BYTEA            | Binary data          | file BYTEA             | BLOB equivalent  |
| Identifier     | UUID             | Unique ID            | id UUID                | Needs extension  |
| Collection     | ARRAY            | Array type           | tags TEXT[]            |                  |
| Spatial        | GEOMETRY         | Spatial data         | location GEOMETRY      | PostGIS          |

```
CREATE TABLE example_table (  
  col1 INTEGER,  
  col2 SMALLINT,  
  col3 BIGINT,  
  col4 NUMERIC(10,2),  
  col5 REAL,  
  col6 DOUBLE PRECISION,  
  col7 CHAR(1),  
  col8 VARCHAR(50),  
  col9 TEXT,
```

```
col10 BYTEA,  
col11 BOOLEAN,  
col12 DATE,  
col13 TIMESTAMP,  
col14 TIMESTAMPTZ,  
col15 TIME,  
col16 INTERVAL,  
col17 JSONB,  
col18 UUID  
);
```

## Column Add Remove

```
ALTER TABLE employees  
ADD COLUMN date_of_joining DATE;
```

```
ALTER TABLE employees  
DROP COLUMN address;
```

```
ALTER TABLE employees  
RENAME COLUMN name TO full_name;
```

## Column Behavior And Properties

```
CREATE TABLE example_table (  
  
    id BIGSERIAL PRIMARY KEY,  
  
    -- Integer type with NOT NULL  
    col1 BIGINT NOT NULL,  
  
    -- Financial data with default value  
    col2 NUMERIC(10,2) DEFAULT 0.00,  
  
    -- Floating value with CHECK constraint  
    col3 DOUBLE PRECISION CHECK (col3 >= 0),  
  
    -- Text with NOT NULL  
    col4 TEXT NOT NULL,  
  
    -- Boolean with default  
    col5 BOOLEAN DEFAULT FALSE,  
  
    -- Timestamp with auto current time  
    col6 TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    -- JSON with default empty object  
    col7 JSONB DEFAULT '{} '::jsonb,  
  
    -- Array with default empty array  
    col8 TEXT[] DEFAULT ARRAY[] ::TEXT[],  
  
    -- UUID with auto generation  
    col9 UUID DEFAULT gen_random_uuid() UNIQUE  
);
```

| Property Name | Short Description                    | Use Case                                    |
|---------------|--------------------------------------|---------------------------------------------|
| PRIMARY KEY   | Unique identifier + NOT NULL + Index | Table এর unique row identify করতে (ex: id ) |
| NOT NULL      | Column এ value অবশ্যই থাকতে হবে      | Required field (ex: name , email )          |
| NULL          | Value optional                       | Optional field (ex: middle_name )           |

| Property Name                     | Short Description          | Use Case                         |
|-----------------------------------|----------------------------|----------------------------------|
| DEFAULT                           | Value না দিলে auto set হয় | Status, timestamps, flags        |
| UNIQUE                            | Duplicate value allow না   | Email, username                  |
| CHECK                             | Custom validation rule     | Positive amount, age limit       |
| FOREIGN KEY                       | অন্য table reference       | Relation (user_id → users table) |
| INDEX                             | Query speed বাড়ায়        | Frequently searched column       |
| AUTO INCREMENT (SERIAL/BIGSERIAL) | Automatic number generate  | ID column                        |
| CURRENT_TIMESTAMP                 | Current date time set      | created_at column                |
| GENERATED ALWAYS AS               | Computed column            | Calculated field (total price)   |
| COMMENT                           | Column documentation       | Schema documentation             |
| COLLATE                           | Text sorting rule          | Language specific sorting        |
| ARRAY DEFAULT                     | Empty array set            | Tags list                        |
| JSON DEFAULT                      | Default JSON value         | Metadata storage                 |

```
CREATE TABLE app_users (  
  
    -- PRIMARY KEY + AUTO INCREMENT  
    id BIGSERIAL PRIMARY KEY,  
  
    -- Required field  
    full_name VARCHAR(150) NOT NULL,  
  
    -- UNIQUE constraint  
    email VARCHAR(150) UNIQUE NOT NULL,  
  
    -- Optional field (NULL allowed)  
    phone VARCHAR(20),  
  
    -- CHECK constraint  
    age INT CHECK (age >= 18),  
  
    -- DEFAULT value  
    is_active BOOLEAN DEFAULT TRUE,  
  
    -- Timestamp default  
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  
    -- JSON metadata  
    profile JSONB DEFAULT '{}'::jsonb,  
  
    -- Array column  
    tags TEXT[] DEFAULT ARRAY[]::TEXT[],  
  
    -- UUID auto generated  
    public_id UUID DEFAULT gen_random_uuid() UNIQUE,  
);
```